Intensity, decibels and Doppler

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish intensity change from frequency shift.

SUCCESS CRITERIA - I can...

1. Use an intensity ratio separately from pitch.
2. Interpret a logarithmic sound-level difference.
3. Predict approach and recession Doppler trends.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “intensity” and “decibel” to compare your model or result. Sentence starters are optional.

intensity: power passing through a unit area

decibel: a unit for a logarithmic sound-level comparison

Doppler shift: a frequency change associated with relative motion

01 Warm-up: bring back an earlier idea

A 500 Hz tone lasts 0.3 s.
Find number of cycles and period.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

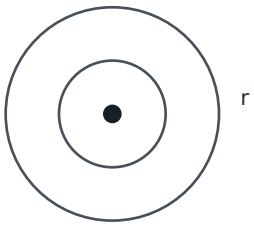
02 Worked example: explain the model

Reference: $\Delta\beta = 10 \log_{10}(I_2/I_1)$.
I is intensity; β is sound level in dB. Tenfold intensity gives +10 dB. Approach generally raises observed frequency; recession lowers it.

Worked reasoning: For the same point source, doubling distance spreads its power over four times the area, so ideal intensity drops to one quarter. This assumes spherical spreading without extra absorption.

Your turn: Draw two concentric wavefronts and indicate why the outer surface shares the power more widely.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Label distances and compare intensity.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

A moving source approaches a stationary listener, then recedes.
Sketch closer wavefront spacing in front and wider spacing behind; compare heard pitch.

Sentence starter: Both … . They differ in … because … .

Source moves right. Add wavefronts.



Marzano check - circle: ① ② ③

04 Apply the physics

A sound meter records an intensity increase by a factor of 4.
Use $\Delta\beta = 10 \log_{10}(I_2/I_1)$ to find the level change. Then compare the intensities of 60 dB and 80 dB sounds. Complete the ratio table.

Sentence starter: Both … . They differ in … because … .

Case	Intensity ratio	Level change (dB)
A	____	____
B	____	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A loud stationary siren and a quiet approaching siren are compared. Does “louder” alone mean “higher frequency”?
Explain 0.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 4 kg sample with c=500 J/(kg·°C) warms 6°C.
Find absorbed energy. If it is 5% of source energy, find source energy.
Label energy entering, leaving and stored.

Word help: absorbed = taken into the specified object or system

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

Stationary listener; source approaches at 20 m/s; emits 500 Hz; v_{sound}=340 m/s.
Use $f_{obs}=f_s\ v/(v-v_s)$.
Find observed frequency.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Source moves right. Add wavefronts.



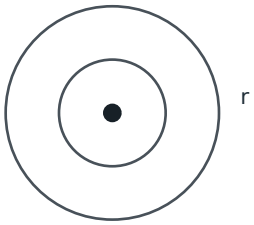
Marzano check - circle: 1 2 3

08 STRETCH: test the model

Using $\Delta\beta=10\ \log_{10}(I_2/I_1)$, find the level change when distance triples from an ideal point source. Separately explain the higher observed pitch of an approaching source.

Word help: ideal = using stated simplifying conditions

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Label distances and compare intensity.

Marzano check - circle: 1 2 3